

Multiple Choice (10 marks)

1. What is the output of "abc"[-1] in Python 3?
 - (a) Error
 - (b) a
 - (c) b
 - (d) c
2. Let l = [1,2,3,4]. What is sum(l) in Python 3?
 - (a) 4
 - (b) 1
 - (c) 10
 - (d) 8
3. Which of the following is NOT a reasonable justification for choosing to busy-wait on an asynchronous event?
 - (a) The wait is expected to be short.
 - (b) A busy-wait loop is easier to code than an interrupt handler.
 - (c) There is no other work for the processor to do.
 - (d) The program executes on a time-sharing system.
4. Which of the following sets of bit strings CANNOT be described with a regular expression?
 - (a) All bit strings whose number of zeros is a multiple of five
 - (b) All bit strings starting with a zero and ending with a one
 - (c) All bit strings with an even number of zeros
 - (d) All bit strings with more ones than zeros
5. What is the output of 4*1**4 in python?
 - (a) 4
 - (b) 1
 - (c) 256
 - (d) 16

True / False (10 marks)

Answer True or False

6. True or False: The number of distinct functions mapping a finite set A with m elements into a finite set B with n elements is $n!/(n - m)!$
7. True or False: $O(n^2)$ grows faster than $O(n)$ and $O(\log n)$ as the input size increases.
8. True or False: Local caching of files ensures that the file system remains entirely consistent across all machines at all times.
9. True or False: Priority queuing is a starvation-free job-scheduling policy that guarantees all jobs receive equal service over time.
10. True or False: The function `random()` in Python 3 sets the integer starting value used in generating random numbers.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a python function to find the maximum occurring character in a given string.
12. Write a function to multiply two lists using map and lambda function.

End of Paper